

RESEARCH ARTICLE

A data-driven T_2 relaxation analysis approach for myelin water imaging: Spectrum analysis for multiple exponentials via experimental condition oriented simulation (SAME-ECOS)

Hanwen Liu^{1,2}  | Tigris S. Joseph^{1,2} | Qing-San Xiang^{1,3} | Roger Tam^{3,4} | Piotr Kozłowski^{2,3}  | David K. B. Li³ | Alex L. MacKay^{1,3} | John L. K. Kramer^{2,5} | Cornelia Laule^{1,2,3,6} 

¹Physics & Astronomy, University of British Columbia, Vancouver, British Columbia, Canada

²International Collaboration on Repair Discoveries, University of British Columbia, Vancouver, British Columbia, Canada

³Radiology, University of British Columbia, Vancouver, British Columbia, Canada

⁴Biomedical Engineering, University of British Columbia, Vancouver, British Columbia, Canada

⁵Kinesiology, University of British Columbia, Vancouver, British Columbia, Canada

⁶Pathology & Laboratory Medicine, University of British Columbia, Vancouver, British Columbia, Canada

Correspondence

Hanwen Liu, Physics & Astronomy,
University of British Columbia, 818
W 10th ave, Vancouver, BC V5Z 1M9,
Canada.

Email: lhwsci@gmail.com

Funding information

Multiple Sclerosis Society of Canada;
Natural Sciences and Engineering
Research Council of Canada;
International Collaboration on Repair
Discoveries

Purpose: The decomposition of multi-exponential decay data into a T_2 spectrum poses substantial challenges for conventional fitting algorithms, including non-negative least squares (NNLS). Based on a combination of the resolution limit constraint and machine learning neural network algorithm, a data-driven and highly tailorable analysis method named spectrum analysis for multiple exponentials via experimental condition oriented simulation (SAME-ECOS) was proposed.

Theory and Methods: The theory of SAME-ECOS was derived. Then, a paradigm was presented to demonstrate the SAME-ECOS workflow, consisting of a series of calculation, simulation, and model training operations. The performance of the trained SAME-ECOS model was evaluated using simulations and six in vivo brain datasets. The code is available at <https://github.com/hanwenat/SAME-ECOS>.

Results: Using NNLS as the baseline, SAME-ECOS achieved over 15% higher overall cosine similarity scores in producing the T_2 spectrum, and more than 10% lower mean absolute error in calculating the myelin water fraction (MWF), as well as demonstrated better robustness to noise in the simulation tests. Applying to in vivo data, MWF from SAME-ECOS and NNLS was highly correlated among all study participants. However, a distinct separation of the myelin water peak and the intra/extra-cellular water peak was only observed in the mean T_2 spectra determined using SAME-ECOS. In terms of data processing speed, SAME-ECOS

is approximately 30 times faster than NNLS, achieving a whole-brain analysis in 3 min.

Conclusion: Compared with NNLS, the SAME-ECOS method yields much more reliable T_2 spectra in a dramatically shorter time, increasing the feasibility of multi-component T_2 decay analysis in clinical settings.

KEYWORDS

data-driven, machine learning, myelin water imaging, non-negative least squares, resolution limit, SAME-ECOS

1 | INTRODUCTION

T_2 relaxation in biological tissues measured with a multi-echo experiment is typically characterized by multi-exponential decays because several water pools may exist within a single image voxel.¹ The T_2 times of different water pools are governed by the microenvironment of water molecules. For example, myelin water, the water trapped in myelin bilayers, exhibits a shorter T_2 relaxation time than that of intra/extra-cellular water (IE) and free water.² Researchers have used the myelin water information as a myelin biomarker to study many myelin-related diseases in the brain and spinal cord, including multiple sclerosis,^{3,4} schizophrenia,⁵ Parkinson disease,⁶ stroke,⁷ concussion,⁸ and cervical spondylotic myelopathy.⁹ Beyond the use of myelin water in neurological conditions, there are other applications of quantitative T_2 assessment, such as in knee¹⁰ and prostate diseases,¹¹ which also rely on extracting information about a specific T_2 component from the multi-echo relaxation experiment.

However, accurately depicting the spectrum of constituent T_2 components for each image voxel from MR relaxation data is nontrivial. Because the decay process of multiple water pools takes place simultaneously, the MR receiver coil can only record a signal that is the sum of multiple exponential decay components. Consequently, one has to solve a mathematical problem of fitting a superimposed relaxation signal into its constituent components. This mathematically complex problem is commonly seen in many other quantitative sciences and is often considered as an ill-posed problem.¹² The analytical and numerical solutions to this problem have been comprehensively reviewed by Istratov et al from a mathematical perspective.¹²

To provide a suitable solution specific to MR data, Whittall and Mackay¹³ introduced the non-negative least squares (NNLS)¹⁴⁻¹⁶ method that decomposes the multi-echo decay data into a spectrum of positive T_2 times. Later, Prasloski et al¹⁷ improved the NNLS method of MR data fitting by applying the extended phase graph (EPG) algorithm to correct for stimulated echoes but this addition

resulted in very long processing times (hours). More recently, Yoo et al¹⁸ and Doucette et al¹⁹ reduced the NNLS whole-brain analysis time to several minutes by using advanced computational hardware and software. The NNLS method makes no prior assumptions about the total number of T_2 components, which is a desirable feature for modeling complex biological tissues with heterogeneous compositions. However, without constraining the number of components, the T_2 fitting problem becomes underdetermined with non-unique solutions, making NNLS unstable and highly susceptible to noise even with strong regularization. In contrast, other fitting methods, such as the quasi-Newton algorithm by Du et al²⁰ and the Wald distribution by Akhondi-Asl et al,²¹ usually provide more stable results and better noise resistance, but at the expense of modelling only two or three water pools, which limits their usage for unknown pathological tissues.

Simultaneously increasing the complexity and stability of a model is a paradox, seemingly impossible to improve one without diminishing the other in the game of multi-exponential decomposition. Our quest now was to find the line of best balance between these two factors. According to previous theoretical studies, the decay components can only be resolved to a certain resolution limit at a given signal-to-noise ratio (SNR).^{22,23} That is to say, due to noise contamination, there is always a limit in how closely the two neighboring components can be resolved, regardless of how sophisticated the analysis method is. This fundamental restriction on the resolution limit leads to a correlation between the SNR and the maximum number of components that can be reliably resolved in a particular analysis range.²⁴ The exact expression of this correlation is derived in the Theory section and plays a crucial role in our proposed approach.

In recent years, machine learning (ML) algorithms have been successfully implemented in many MR applications.²⁵ Particularly regarding the multi-component T_2 fitting, Lee et al²⁶ and Liu et al²⁷ applied neural network algorithms²⁸ to replicate the NNLS calculation, and achieve whole brain analysis in seconds. Nagtegaal et al²⁹ and Kumar et al³⁰ proposed to use a joint sparsity constraint

and a 3D spatial regularization, respectively, to boost the spatial smoothness of the NNLS results. Does et al³¹ used principal component analysis denoising to improve the NNLS fitting stability. However, most of these ML-based methods implemented either spatial regularization or denoising techniques on top of NNLS, leaving the core NNLS algorithm untouched.

Building upon these well-established ML methods, we propose a data-driven method that incorporates the T_2 resolution limit as a constraint when computing the solution of a T_2 spectrum. Our proposed method is called Spectrum Analysis for Multi-Exponentials via Experimental Condition Oriented Simulation (SAME-ECOS) for the analysis of multi-echo T_2 relaxation data. The general concept of SAME-ECOS (Figure 1) can be briefly described by a series of calculation, simulation, and model training operations:

1. Determine the T_2 range and resolution limit based on the experimental conditions such as SNR and echo times;
2. Generate sufficient examples of random T_2 spectra within the T_2 range obeying the resolution limit;
3. Compute multi-echo decay data using the randomly generated T_2 spectra;
4. Train a ML neural network model to learn the mapping between the simulated multi-echo decay data and the ground truth spectrum labels;
5. Apply the trained neural network model to experimental data.

In general, SAME-ECOS is a data-driven solver, constrained by resolution limit, to the problem of fitting multi-exponential decay data into a T_2 spectrum. It is worth highlighting that SAME-ECOS has high flexibility of tailoring itself to different experimental conditions attributed to its unique simulation workflow. The detailed SAME-ECOS algorithm is demonstrated in the Methods by presenting an example with in-depth explanations.

2 | THEORY

The theoretical foundation of SAME-ECOS was built upon previous studies²²⁻²⁴ with some further development. This section presents the mathematical derivation of the relationship between the experimental SNR and the resolvable T_2 components in a multi-exponential decomposition problem.

In the time domain, the evolution of the MR signal in a multi-echo experiment can be expressed as

$$f(t) = \int_0^{\infty} e^{-\gamma t} g(\gamma) d\gamma \text{ with } t \geq 0 \text{ and } \gamma \geq 0 \quad (1)$$

where the $g(\gamma)$ is the function of the T_2 spectrum in linear scale. Equation (1) can be transformed to logarithmic scale by letting $x = \ln t$ and $y = -\ln \gamma$ to have

$$f(e^x) = \int_{-\infty}^{+\infty} e^{-e^{x-y}} g(e^{-y}) e^{-y} dy \quad (2)$$

where the right-hand side is the convolution of two functions e^{-e^x} and $g(e^{-y})e^{-y}$. Applying deconvolution to Equation (2) and solving for $g(e^{-y})$ takes the form of

$$g(e^{-y}) = F^{-1} \left\{ \frac{F\{f(e^x)\}}{F\{e^{-e^x}\}} \right\} e^y \quad (3)$$

where F and F^{-1} denotes the Fourier and inverse Fourier transformation. In the case of discrete $g(\gamma) = \sum_n a_n \delta(\gamma - \gamma_n)$ and $f(t) = \sum_n a_n e^{-\gamma_n t} + v(t)$, where $v(t)$ is the noise term, then the signal in the logarithmic scale becomes $f(e^x) = \sum_n a_n e^{-e^{x-y_n}} + v(e^x)$. Subsequently, Equation (3) can be written as

$$g(e^{-y}) = F^{-1} \left\{ \frac{\sum_n a_n e^{i\omega y_n} F\{e^{-e^x}\} + F\{v(e^x)\}}{F\{e^{-e^x}\}} \right\} e^y \quad (4)$$

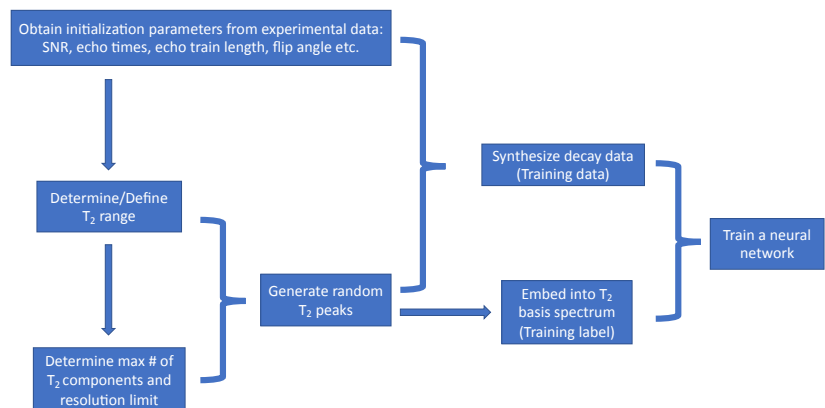


FIGURE 1 SAME-ECOS workflow

Let $e^x = \theta$, then $F\{e^{-e^x}\} = \int_{-\infty}^{+\infty} e^{-e^x} e^{-i\omega x} dx = \int_0^{+\infty} e^{-\theta} \theta^{\theta-1} d\theta = \Gamma(i\omega)$, so the T_2 spectrum function $g(e^{-y})$ is solved as

$$g(e^{-y}) = F^{-1} \left\{ \frac{\sum_n a_n e^{i\omega y_n} \Gamma(i\omega) + F\{v(e^x)\}}{\Gamma(i\omega)} \right\} e^y \quad (5)$$

Equation (5) indicates that the solution is band limited in the frequency domain, meaning only those frequency components that are larger than the noise term can be properly reconstructed in the inverse Fourier transformation. Therefore, a maximum frequency $\omega_{\max}$ exists and satisfies

$$a_n^2 |\Gamma(i\omega_{\max})|^2 = |F\{v(e^x)\}|^2 \quad (6)$$

where the phase factor $e^{i\omega y_n}$ is demodulated out. Using a known property of the gamma function

$$|\Gamma(i\omega)|^2 = \frac{\pi}{\omega \sinh(\pi\omega)} \quad (7)$$

and assuming a constant white noise with amplitude v_0 , the maximum frequency can be determined by

$$\frac{\omega_{\max}}{\pi} \sinh(\pi\omega_{\max}) = \left(\frac{a_n}{v_0}\right)^2 \quad (8)$$

If the T_2 spectrum consists of M components with equal amplitudes a_0 , then $S_0 = f(t=0)$, $a_0 = S_0/M$, and $SNR = S_0/v_0$ to give

$$\frac{\omega_{\max}}{\pi} \sinh(\pi\omega_{\max}) = \left(\frac{SNR}{M}\right)^2 \quad (9)$$

Since $\omega_{\max}$ is the most oscillatory frequency component to construct the T_2 spectrum in logarithmic scale, any two adjacent T_2 components (denoted as y_n and y_{n-1}) must have a minimum separation to be resolvable

$$y_n - y_{n-1} = \frac{\pi}{\omega_{\max}} \quad (10)$$

In linear scale

$$\ln \gamma_n - \ln \gamma_{n-1} = \frac{\pi}{\omega_{\max}} \quad (11)$$

Then, the resolution limit δ takes the form of

$$\delta = \frac{\gamma_n}{\gamma_{n-1}} = e^{\pi/\omega_{\max}} \quad (12)$$

Alternatively, if the T_2 spectrum can be resolved up to M components in the range $[T_2^{\min}, T_2^{\max}]$, then the resolution limit can also be expressed as

$$\delta = \left(\frac{T_2^{\max}}{T_2^{\min}} \right)^{1/M} \quad (13)$$

Combining Equations (9), (12), and (13) one can obtain an expression that correlates the SNR, T_2 range, and M resolvable components:

$$\frac{M}{\ln\left(\frac{T_2^{\max}}{T_2^{\min}}\right)} \times \sinh\left(\frac{\pi^2 \times M}{\ln\left(\frac{T_2^{\max}}{T_2^{\min}}\right)}\right) = \left(\frac{SNR}{M}\right)^2 \quad (14)$$

Note that, the above theoretical derivation of SAME-ECOS should be universally applicable to any multi-exponential decay signals. However, for the interest of the MR community, this paper is only limited to applying the SAME-ECOS method to myelin water imaging data.

3 | METHODS

Because the SAME-ECOS analysis method is tailored to different MR experimental conditions, we use one specific example here to demonstrate its simulation workflow, trained model evaluation, and experimental data application.

3.1 | In vivo MRI experiment

The 32-echo brain data (gradient and spin echo (GRASE), TE/ Δ TE/TR = 10/10/1000 ms, refocusing flip angle (FA) = 180°, axial matrix size = 232 × 186, acquired resolution = 1 × 1 × 5 mm³ for 20 slices, reconstructed resolution = 1 × 1 × 2.5 mm³ for 40 slices, acquisition time = 14.4 min)³² from six healthy volunteers (six males, age range: 29-70 y old, mean age: 46 y old) were collected at 3T (Philips Achieva) using an eight-channel head coil. The volunteers provided signed informed consent before participating under the approval of the University of British Columbia Clinical Research Ethics Board.

3.2 | SAME-ECOS workflow

1. **Choose the SNR range to be 50-500.** The chosen SNR range should be substantially large to cover all possible SNRs for the in vivo data acquisition.

2. **Define the T_2 range to be 5-2000 ms.** The upper boundary $T_2^{\max} = 2000$ ms was chosen empirically,^{1,2,15} and the lower boundary $T_2^{\min} = 4.827$ ms ≈ 5 ms was determined at $SNR = 500$ and the 3rd echo time = 30 ms by

$$T_2^{\min} = -\frac{3rd \text{ echo time}}{\ln\left(\frac{1}{SNR}\right)} \quad (15)$$

Equation (15) requires the residual signal of the shortest T_2 component to be higher than the noise level at the third echo.

3. **For each simulation realization:**

- generate a random integer $SNR \in [50, 500]$.** The SNR was sampled from a uniform distribution.
- determine the maximum number of resolvable T_2 components M and the resolution limit δ .** M and δ was calculated using Equations (13) and (14) with the defined T_2 range at the selected SNR. For example, $M = 5$ and $\delta = 3.314$ when $SNR = 500$.
- generate random integers $n < M$, $FA \in [90^\circ, 180^\circ]$.** Note that the simulation did not include the case of $n = M$. Our post hoc analysis showed that when $n = M$, certain T_2 locations and amplitudes would get generated more frequently in the next step, introducing unwanted biases to the simulation of multiple T_2 components. Also note that the actual FA can deviate substantially from the prescribed FA due to B1 inhomogeneity, so we gave a 90° tolerance to account for the FA variations.
- generate n T_2 components within the T_2 range.** The randomly generated n T_2 locations should obey the resolution limit δ . When $n \geq 2$, the maximum frequency $\omega_{\max}$ can be found using Equation (11) and substituted into Equation (8) to calculate the lowest detectable amplitude a_n . Then, the amplitude of each T_2 component was randomly drawn from a uniform distribution within the range $[a_n, 1 - na_n]$. The amplitudes of n T_2 components were subsequently normalized (sum to unity).
- synthesize 32-echo decay data.** The decay signal of the n T_2 components were synthesized using the extended phase graph (EPG) algorithm^{17,33} at selected FA with $T_1 = 2000$ ms. Gaussian noise for both real and imaginary parts were added at the selected SNR level, and the magnitude signal was recorded for further processing.³⁴
- embed n T_2 components into a spectrum representation depicted by 40 basis T_2 s.** The 40 basis T_2 s ($t_1, t_2, \dots, t_{40}$) were logarithmically spaced in the T_2

range to represent the spectrum of n T_2 components ($T_{2,1}, T_{2,2}, \dots, T_{2,n}$). The weighting factor w_i of the i^{th} basis $T_2(t_i)$ was calculated by

$$w_i = \sum_{j=1}^n \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}(t_i - T_{2,j})^2} \quad (16)$$

The embedded T_2 basis representation was normalized (sum to unity) and recorded for model training.

4. **Train a neural network to map the decay data to its T_2 spectrum.** A neural network (hidden layers: $100 \times 500 \times 1000 \times 1000 \times 500$, activation: SeLU³⁵ (hidden layers) and softmax³⁶ (output layer), optimizer: Adamax,³⁷ loss: categorical cross-entropy^{37,38}) was constructed using TensorFlow³⁹ to take 32-echo decay data as input and predict the weighting factors of the 40 basis T_2 s at the output layer. The constructed neural network was trained to map the decay data to its T_2 spectrum. We yielded 3,000,000 simulation realizations (step 3 above), 80% of which were used for the neural network training. The remaining 20% simulation realizations were used for the validation that determined the stopping criterion for the training process. The training was stopped when the accuracy on the validation set did not improve further. This trained neural network is denoted as the SAME-ECOS model hereafter and can be applied to new 32-echo decay data to obtain T_2 spectrum.

3.3 | SAME-ECOS model performance evaluation

The performance of the SAME-ECOS model was evaluated using three tests and compared respectively with the results determined by a regularized NNLS solver equipped with stimulated echo correction²⁴ (analysis program can be requested from <https://mrresearch.med.ubc.ca/news-projects/myelin-water-fraction/>). The kernel matrix for the NNLS analysis was adjusted accordingly to match our experimental and simulation parameters. The regularization parameter was chosen to be the largest value that allows a misfit of less than 1.02 times the minimum misfit, which is commonly used in many studies.⁴⁰⁻⁴⁴

Test 1: Four Monte Carlo phantoms were simulated to visualize the spectrum prediction performance when there is $n = 1, 2, 3$, and 4 T_2 components. Each phantom has a dimension of 91×100 voxels. The phantom width (91) from left to right corresponds to the SNR ranging from 50 to 500 with an increasing step size of 5. At each SNR, 100 ground truth T_2 spectra were randomly generated, and the

32-echo decay data were subsequently produced for each voxel, following steps described in Section 3.2. The decay data of these phantoms were analyzed by the SAME-ECOS model and NNLS, respectively, to produce the T_2 spectra for each voxel. The 'goodness' of each spectrum fitting by both methods was quantitatively assessed using cosine similarity scores,⁴⁵ which report values between 0 to 1, with 0 being the least similar and 1 being the most similar to the ground truth labels. The cosine similarity score is a commonly used metric that measures the similarities between two vectors, especially when the vectors are high dimensional. It is a suitable metric for our task since each T_2 spectrum can be treated as a vector of 40 dimensions. The calculation of the cosine similarity score is defined explicitly as

$$\text{cosine similarity score} = \frac{X \cdot Y}{\|X\| \times \|Y\|} \quad (17)$$

where X and Y are the vector representations of the predicted and the ground truth spectra; $\|X\|$ and $\|Y\|$ are their Euclidean norms respectively. Mean cosine similarity with 95% confidence level was calculated for each SNR. A paired t-test was performed to determine whether there was a significant difference ($P < .05$) in the cosine similarity scores calculated by SAME-ECOS and NNLS.

Test 2: The myelin water fraction (MWF, the fraction of signal with $T_2s < 40$ ms)² of the three Monte Carlo phantoms ($n = 2, 3, 4$) were exacted from the predicted spectra by both methods and compared with the ground truth MWF. For better visualization, the 100 voxels in the height direction were rearranged in ascending order of the ground truth MWF at each SNR. Mean absolute error (MAE) with 95% confidence level was computed for each SNR. A paired t-test was performed to determine whether there was a significant difference ($P < .05$) in MAE between SAME-ECOS and NNLS.

Test 3: To examine the model robustness to noise, simulated decay data of 4 pre-defined ground truth T_2 spectra (Table 1) at SNR = 200 and FA = 180°, each with 100 different noise realizations, were passed to the SAME-ECOS model and NNLS for spectrum predictions. The location

and amplitude of each ground truth spectrum were manually chosen to provide a visual-friendly data presentation. The spectrum analysis results of both SAME-ECOS and NNLS were normalized (sum to unity) prior to comparison. The similarity between each predicted and ground truth spectrum was assessed by cosine similarity scores defined above. The mean and standard deviation of the cosine similarity scores were also calculated.

3.4 | Apply SAME-ECOS to experimental in vivo data

The SAME-ECOS model was applied to the experimental in vivo GRASE data, which were pre-processed by normalizing to the first echo image. The processing time for the whole brain data analysis was recorded. From the resulting T_2 spectra, the MWF ($T_2s < 40$ ms) was extracted for each voxel. The masks of regions of interest (ROIs), including segmented white matter, corpus callosum, corticospinal tract, forceps major, and forceps minor, were produced using the first echo image via the FSL segmentation tool.⁴⁶ The T_2 spectra and MWF map was also produced using NNLS in the T_2 range of 5-2000 ms as a reference for comparison. Paired t-tests and Pearson correlations were used to compare SAME-ECOS and NNLS MWF from the 6 volunteers ($P < .05$).

To further validate the robustness to noise, particularly in handling in vivo data, simulated MR noises (SNR = 100) were added to the brain decay data, and the noise-contaminated MWF maps were produced by the SAME-ECOS model and NNLS. The noise addition experiments were performed five times to assess the stability of both methods. The mean and standard deviation of MWF MAE were evaluated. A paired t-test was performed to determine whether there was a significant difference ($P < .05$) in MAE between SAME-ECOS and NNLS.

4 | RESULTS

4.1 | Performance evaluation via simulation tests

In Test 1, the cosine similarity scores of the spectrum predicted by SAME-ECOS were significantly higher (all $P < .05$) than by NNLS in all four Monte Carlo phantoms (Figure 2). For both methods, the similarity scores were increasing with increased SNR from 50 to 500, and the overall performances were better with fewer T_2 components.

In Test 2, SAME-ECOS was able to make more accurate MWF predictions than NNLS with significantly lower MAEs (all $P < .05$) at all SNRs for all three phantoms

TABLE 1 Four pre-defined ground truth spectra

Pre-defined spectra	T_2 locations (ms)	T_2 amplitudes (normalized)
Spectrum 1	100	1
Spectrum 2	25, 120	0.3, 0.7
Spectrum 3	15, 80, 500	0.35, 0.5, 0.15
Spectrum 4	10, 50, 200, 1200	0.35, 0.35, 0.15, 0.15

Note: The T_2 locations of each spectrum are selected obeying the resolution limit. Amplitudes are normalized to one.

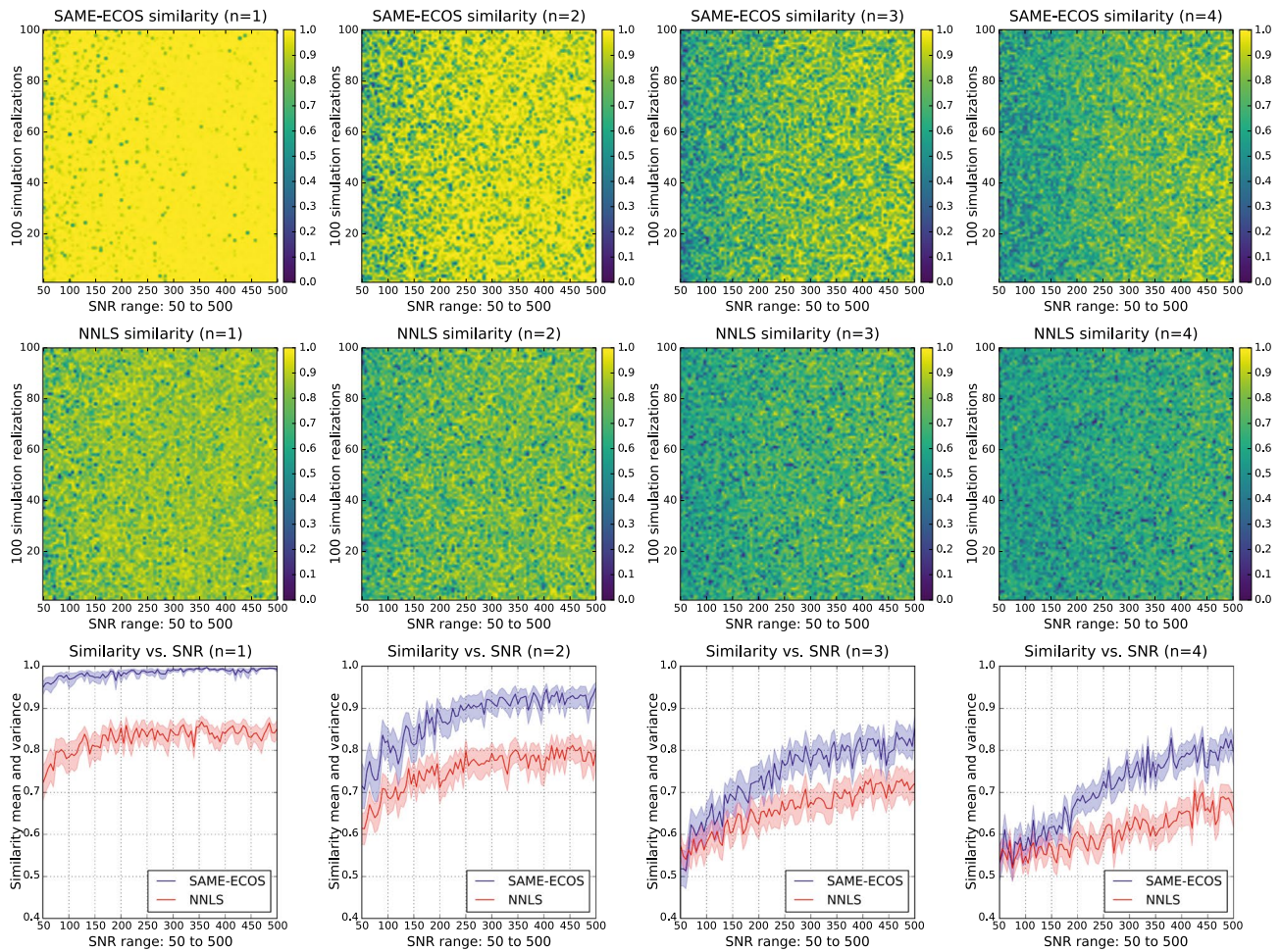


FIGURE 2 SAME-ECOS and NNLS cosine similarity maps of simulated phantoms. Four Monte Carlo phantoms ($n = 1, 2, 3$, and $4 T_2$ components), each with a dimension of 91×100 voxels, were simulated to visualize the spectrum prediction performance. The voxel-wise cosine similarity maps are shown for each phantom (top row: SAME-ECOS, middle row: NNLS). Mean SAME-ECOS (blue) and NNLS (red) cosine similarity scores with 95% confidence levels are plotted against SNR (bottom row)

(Figure 3). MWF MAEs of both methods were decreasing with increased SNR, and MWF calculations were more accurate when there were fewer T_2 components. From a visual inspection, the NNLS MWF maps were generally noisier than the SAME-ECOS MWF maps.

In Test 3, the resulting SAME-ECOS and NNLS spectra from each individual noise realization (grey) were plotted in Figure 4 to compare against the ground truth spectrum (green). The average spectrum from 100 different noise realizations was also calculated (blue: SAME-ECOS, red: NNLS). SAME-ECOS produced significantly better results than NNLS for all scenarios (all $P < .05$). For spectra consisting of one or two T_2 components, the SAME-ECOS model was able to make almost perfect predictions (cosine similarity score: 0.999 ± 0.001 and 0.978 ± 0.027 , respectively); NNLS could also make accurate predictions but with more substantial variability (cosine similarity score: 0.886 ± 0.021 and 0.850 ± 0.062 , for one and two

components, respectively). For spectra consisting of three or four T_2 components, the performances of both methods started to degrade. However, SAME-ECOS (cosine similarity score: 0.835 ± 0.118 , 0.681 ± 0.136 , respectively) was still making better predictions than NNLS (cosine similarity score: 0.679 ± 0.151 , 0.619 ± 0.159 , respectively).

4.2 | In vivo experimental data: MWF maps

In vivo GRASE data were analyzed by the SAME-ECOS model and NNLS. The average processing times of the whole brain data were 3 min for the SAME-ECOS model and 86 min for NNLS (CPU: Intel(R) Core(TM) i7-5930K @ 3.5 GHz, 32 GB RAM). Six representative slices of the GRASE first echo, the resulting MWF maps produced by both methods, and the voxel-wise MWF difference map

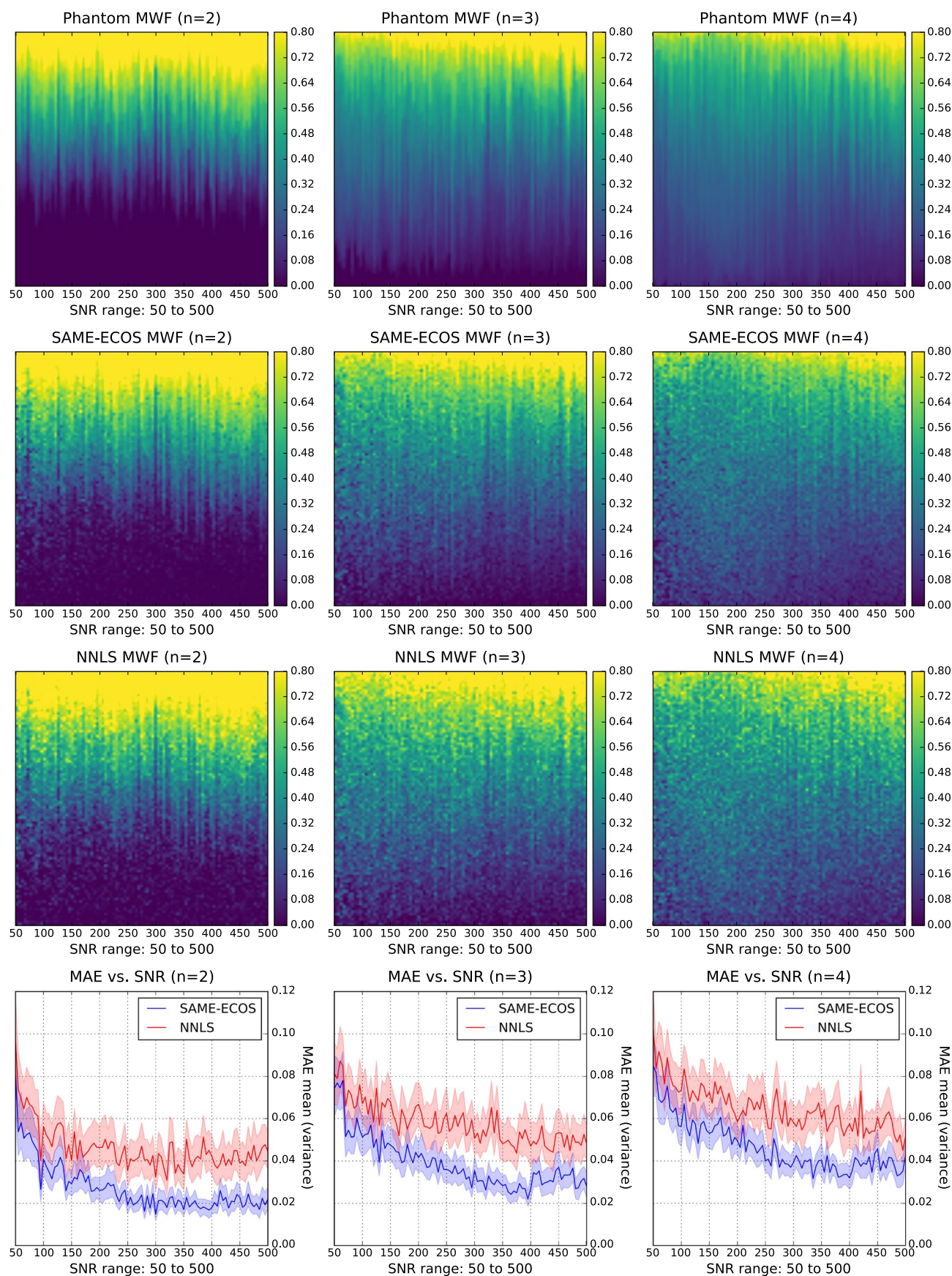


FIGURE 3 SAME-ECOS and NNLS myelin water fraction maps of simulated phantoms. Three Monte Carlo phantoms ($n = 2, 3$, and 4 T_2 components, first row), each with a dimension of 91×100 voxels, were simulated to visualize the performance of MWF ($T_2s < 40$ ms) prediction by SAME-ECOS (second row) and NNLS (third row). Mean absolute error (MAE) of MWF (blue: SAME-ECOS; red: NNLS) with 95% confidence levels are plotted against SNR (fourth row)

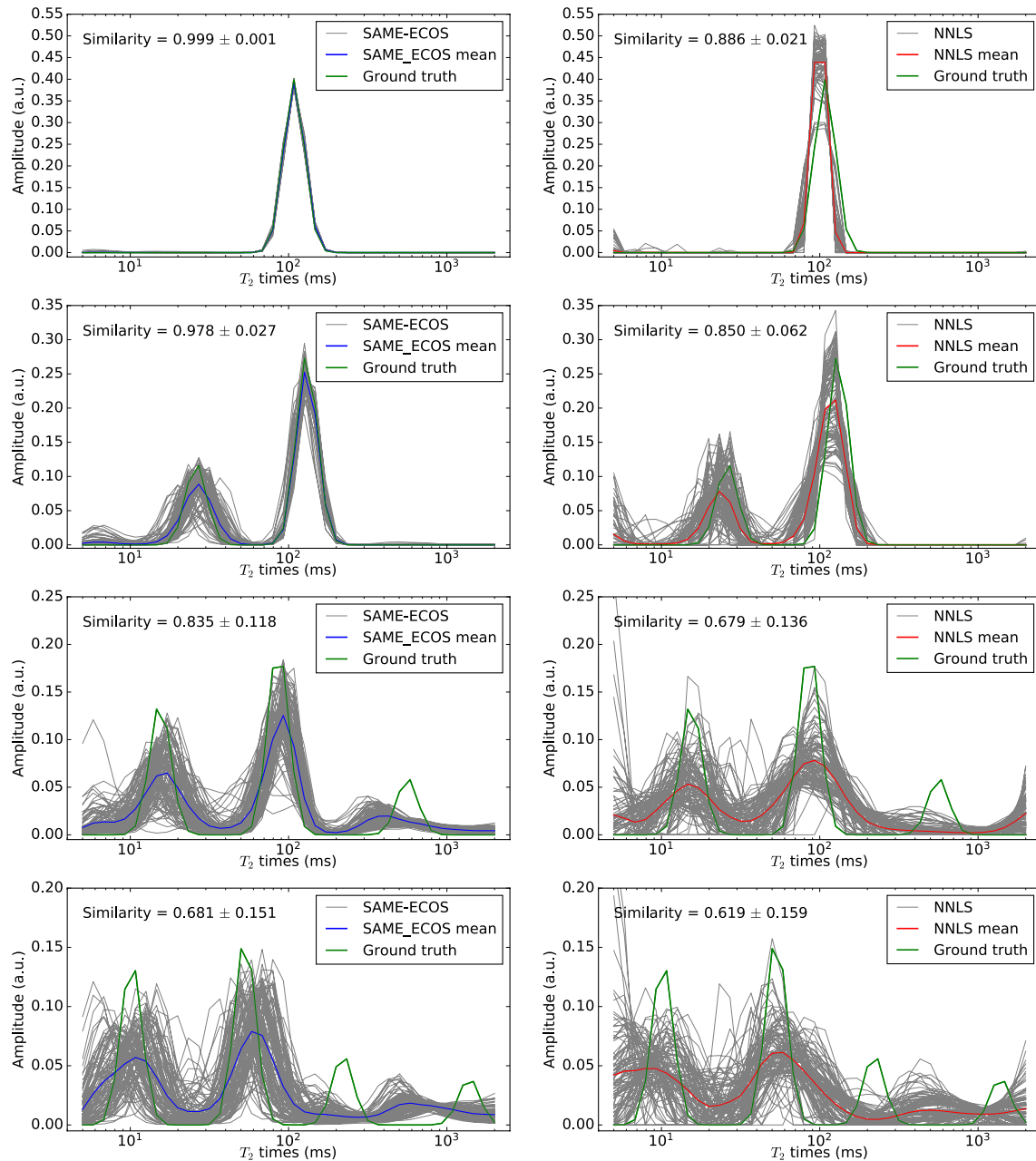


FIGURE 4 T_2 spectra produced by SAME-ECOS (blue) and NNLS (red) respectively are compared with the ground truth spectra (green). Simulated decay data, which are generated from 4 pre-defined ground truth spectra (Table 1) each with 100 different noise realizations, are fed into the trained SAME-ECOS model and NNLS algorithm to generate the T_2 spectra. Faded gray lines indicate the produced spectra for each noise realization. The mean and standard deviation of cosine similarity scores of 100 realizations are shown for each sub-figure

(SAME-ECOS MWF – NNLS MWF) from one participant are presented in Figure 5. The SAME-ECOS MWF map is visually very similar to the NNLS MWF map, but subtle differences are still visible. Figure 6 shows the voxel

spectra, the mean spectra, and the mean MWF within the ROIs of the corpus callosum, corticospinal tract, forceps major, and forceps minor from the example study participant shown in Figure 5. For all ROIs, the SAME-ECOS

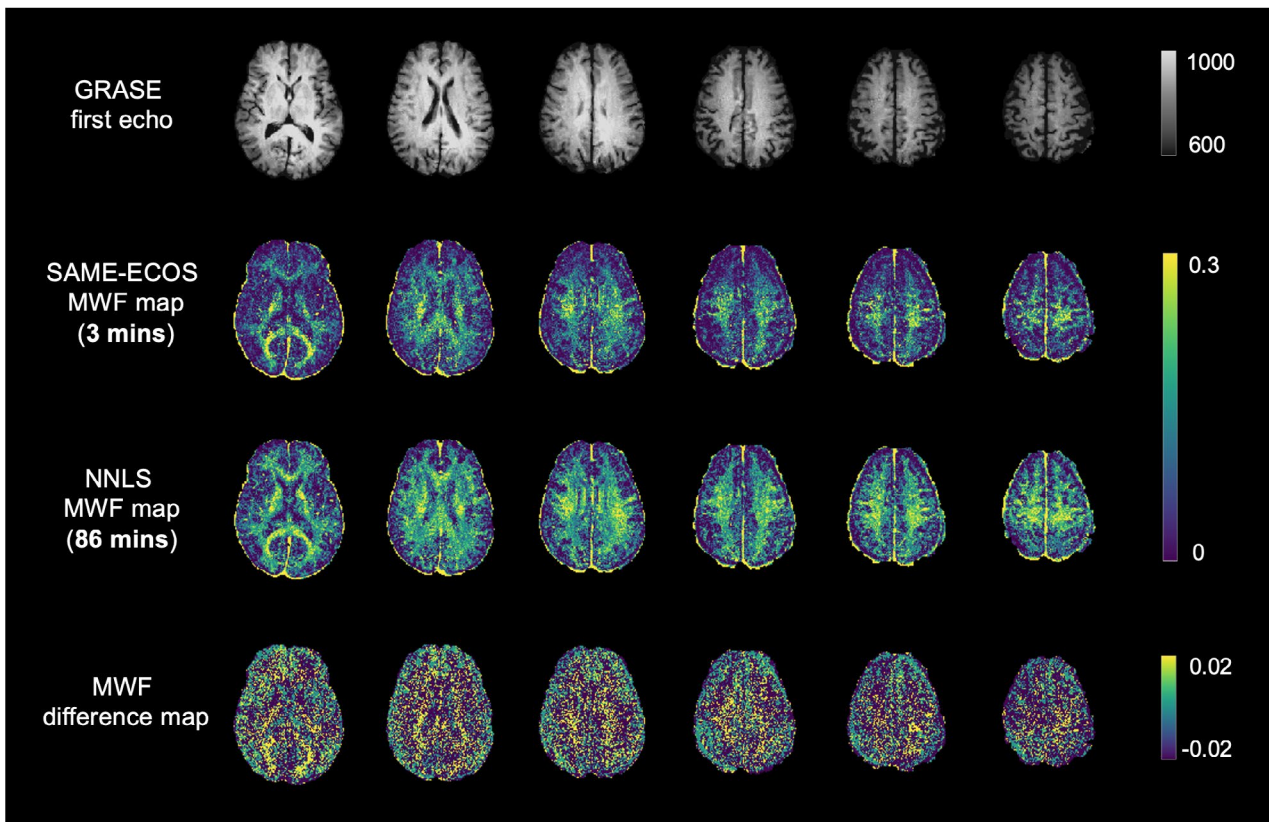


FIGURE 5 SAME-ECOS and NNLS derived in vivo myelin water fraction maps. Six representative slices of the GRASE first echo from one volunteer, the resulting MWF maps produced by both methods ($T_2s < 40$ ms), and the voxel-wise MWF difference map (SAME-ECOS MWF – NNLS MWF), are shown

mean spectra was able to distinctively resolve a short and small T_2 peak (attributed to myelin water) in addition to a more dominant middle peak (attributed to IE water), whereas the NNLS mean spectra could only resolve a single broad middle peak that extends into the myelin water region. Similar observations were found in all study participants. Quantitatively, the SAME-ECOS mean MWF across the 6 participants was on average 10% lower ($P < .05$) (range 0 to -19%) than the NNLS approach (Table 2). SAME-ECOS and NNLS mean MWF were highly correlated in all white matter ROIs (R^2 range 0.848–0.99, $P < .05$) (Table 2).

The effects of additional noise (five trials) are visualized by showing a representative slice of the brain from one study participant in Figure 7. Compared with NNLS, the SAME-ECOS MWF maps produced from the noise-contaminated decay data were visually less noisy and more consistent with their original MWF maps. For all study participants, the mean SAME-ECOS MAE of the five trials with different noise realizations was 16% lower ($P < .05$) than the mean NNLS MAE (Table 3).

5 | DISCUSSION

5.1 | SAME-ECOS vs. NNLS

From the results of all three simulation tests, the SAME-ECOS model largely outperformed NNLS. In the Monte Carlo phantom tests (Tests 1 & 2), the SAME-ECOS model achieved more than 15% higher cosine similarity scores, and over 10% lower MAE of MWF predictions than NNLS at most SNRs (Figures 2 and 3). It was observed that SAME-ECOS was more advantageous over NNLS at higher SNRs. Particularly in the task of resolving three or four T_2 components, only SAME-ECOS could make reliable predictions in the high SNR regime. From visual inspection of the Test 3 results presented in Figure 4, both methods were able to produce accurate T_2 spectra when the number of T_2 components $n \leq 2$. However, when $n > 2$, the NNLS predictions became extremely unstable, resulting in larger variances. This observation illustrates that, unlike NNLS being highly susceptible to noise, the SAME-ECOS model has a desirable feature of being relatively noise-inert. It is also noticed that both methods were incapable

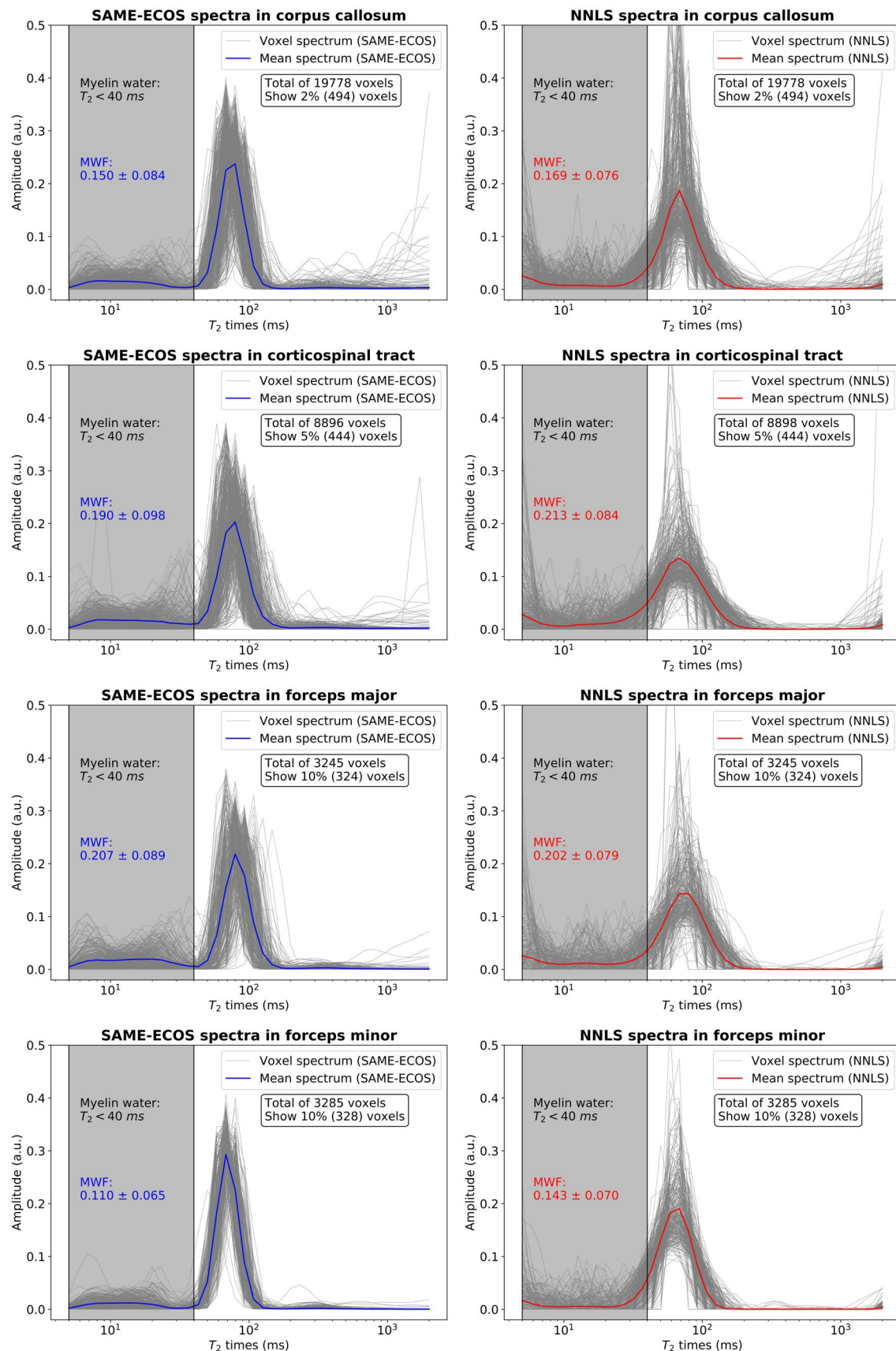


FIGURE 6 Voxel spectra (gray), the mean spectra (blue: SAME-ECOS; red: NNLS), and the mean MWF ($T_2 < 40$ ms) within the four ROIs of the experimental in vivo data. ROIs include the corpus callosum, corticospinal tract, forceps major, and forceps minor. The voxel spectra are only plotted for a fraction of the total number of voxels for a visual-friendly presentation as indicated in each sub-figure. The mean spectrum and mean MWF are calculated from all voxels within each ROI

TABLE 2 Mean MWF in white matter ROIs

MWF in ROIs	White matter	Forceps major	Forceps minor	Corpus callosum	Corticospinal tract
NNLS mean MWF (Std)					
Subject 1	0.146 (0.100)	0.175 (0.067)	0.087 (0.055)	0.144 (0.070)	0.270 (0.131)
Subject 2	0.107 (0.075)	0.141 (0.061)	0.087 (0.063)	0.122 (0.066)	0.200 (0.088)
Subject 3	0.120 (0.093)	0.135 (0.072)	0.078 (0.061)	0.119 (0.070)	0.255 (0.115)
Subject 4	0.174 (0.101)	0.218 (0.092)	0.128 (0.072)	0.172 (0.077)	0.288 (0.121)
Subject 5	0.169 (0.093)	0.189 (0.075)	0.135 (0.064)	0.182 (0.075)	0.271 (0.116)
Subject 6	0.161 (0.084)	0.203 (0.077)	0.131 (0.067)	0.170 (0.072)	0.255 (0.085)
Average	0.146 (0.091)	0.177 (0.074)	0.108 (0.064)	0.152 (0.072)	0.257 (0.109)
SAME-ECOS mean MWF (Std)					
Subject 1	0.119 (0.083)	0.176 (0.071)	0.069 (0.046)	0.126 (0.075)	0.226 (0.091)
Subject 2	0.091 (0.068)	0.147 (0.071)	0.064 (0.050)	0.101 (0.069)	0.190 (0.086)
Subject 3	0.103 (0.077)	0.141 (0.072)	0.067 (0.050)	0.104 (0.066)	0.244 (0.084)
Subject 4	0.152 (0.093)	0.205 (0.082)	0.106 (0.065)	0.151 (0.081)	0.280 (0.094)
Subject 5	0.138 (0.087)	0.191 (0.084)	0.099 (0.060)	0.160 (0.082)	0.242 (0.085)
Subject 6	0.148 (0.082)	0.203 (0.077)	0.123 (0.061)	0.166 (0.076)	0.257 (0.083)
Average	0.125 (0.082)	0.177 (0.076)	0.088 (0.055)	0.135 (0.075)	0.240 (0.087)
Correlation between NNLS and SAME-ECOS					
R ²	0.969	0.990	0.925	0.971	0.848
P-value	0.001	<0.001	0.008	0.001	0.033

Note: The mean MWF (SD) is calculated by NNLS and SAME-ECOS, and the correlation between the two methods is evaluated.

of producing reliable spectrum predictions particularly for the long T_2 components (eg, $T_2 > 500$ ms), which is likely due to the relatively short measurement time (the last TE = 320 ms), so there was not enough information acquired to resolve the long T_2 component accurately. Overall, the current SAME-ECOS model demonstrated a better performance than NNLS in the interrogations using simulation tests.

Due to the lack of ground truth, it is difficult to judge which method is more accurate when it comes to in vivo experimental data. However, in terms of data processing speed, SAME-ECOS is approximately 30 times faster than NNLS, achieving whole-brain analysis in 3 min, which is more feasible in clinical settings. In addition, SAME-ECOS demonstrated better stability in terms of calculating MWF values from noisy in vivo data (Figure 7), potentially mitigating the unwanted within or between-subject variations caused by random machinery noise. Interestingly, from a visual inspection of the MWF maps shown in Figures 5 and 7, these two methods produced similar results, but the SAME-ECOS MWF map appeared to be a little less noisy. A thorough analysis of the NNLS spectra in several white matter ROIs (Figure 6) revealed that the NNLS MWF signal (T_2 cutoff at 40 ms) largely originated from the broad middle peak, which is commonly designated as the IE water pool. On average, there was not much contribution from

a separate myelin water pool ($T_2 < 40$ ms) to the NNLS MWF, and a separate myelin water peak did not exist on the NNLS mean spectra for all ROIs. In contrast, the separation of myelin water and IE water was easily seen on the SAME-ECOS mean spectra for all ROIs (Figure 6), aligning with the intuitive interpretation of MWF arising entirely from a distinct myelin water pool. Thus, this observation suggests that reporting MWF values alone may not fully unveil the underlying information, especially for those myelin water imaging studies that used NNLS. Future studies using histological validation^{47,48} should be conducted to better compare these two analysis methods in dealing with experimental data.

5.2 | Justification for the use of resolution limit

To demonstrate the benefit of applying the resolution limit, the performance of SAME-ECOS with and without considering the resolution limit constraint was compared. SAME-ECOS with the resolution limit followed the steps described in the Methods. For SAME-ECOS without considering the resolution limit, the maximum number of resolvable T_2 components M could not be analytically determined, so an empirical $M = 3$

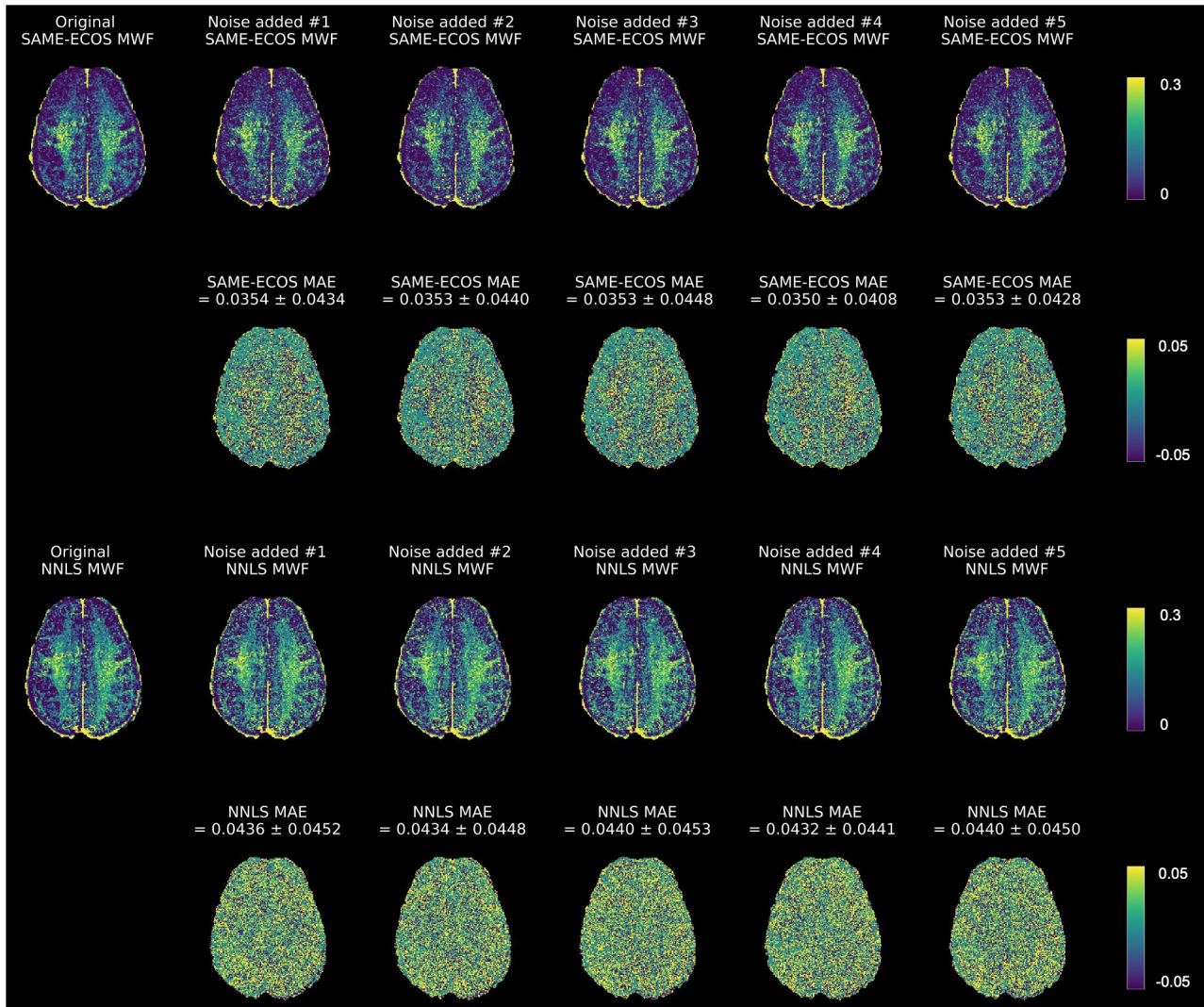


FIGURE 7 In vivo SAME-ECOS and NNLS MWF robustness to noise contamination. Simulated noise at SNR = 100 (5 trials) was added to the original in vivo brain data. The original and noise contaminated MWF maps produced by both methods, and the voxel-wise MWF difference maps (noise added MWF – original MWF), are shown. MAE of MWF and the standard deviations are presented

(myelin water, intra-cellular water, extra-cellular water) was used. Supporting Information Figures S1 and S2, which are available online, show that for the simulation data, SAME-ECOS with resolution limit achieved higher similarity scores and lower MWF MAEs compared to SAME-ECOS without resolution limit. In vivo, SAME-ECOS with resolution limit produced visually less noisy MWF maps (Supporting Information Figure S3) and T_2 spectra with a distinct short, small T_2 peak (attributed to myelin water) in addition to a larger middle peak (attributed to IE water), as opposed a single broad middle peak from SAME-ECOS without resolution limit (Supporting Information Figure S4). Interestingly, SAME-ECOS without resolution limit produced MWF maps and T_2 spectra similar to the NNLS method (Figures 5 and 6). To summarize, these findings illustrate that the application of the resolution limit is necessary and beneficial.

5.3 | Advantages of SAME-ECOS

The biggest advantage of SAME-ECOS lies in the fact that it is a data-driven solver, which adaptively computes the solution for the given MR data. Its unique workflow makes SAME-ECOS highly tailorable to different experimental conditions and the needs of analysis. Most parameters used in SAME-ECOS are tunable. For example, the T_1 parameter was set to be a constant for all T_2 components in our simulation because the T_1 weighting is minimized by using GRASE. But if a sequence sensitive to both T_1 and T_2 were used, one could simply turn the T_1 parameter into another variable for the production of training data and make the training labels a 2D map that resolves distinct components with different T_1 and T_2 times.⁴⁹ In contrast, it is presumably more difficult to include either T_1 or any

TABLE 3 Robustness to added noise

Noise trials	Noise trial 1	Noise trial 2	Noise trial 3	Noise trial 4	Noise trial 5	Mean	Std	CoV
NNLS MWF MAEs								
Subject 1	0.0411	0.0410	0.0410	0.0410	0.0409	0.0410	7.07e-5	0.172%
Subject 2	0.0375	0.0375	0.0374	0.0376	0.0374	0.0375	8.36e-5	0.223%
Subject 3	0.0385	0.0384	0.0384	0.0385	0.0384	0.0384	5.48e-5	0.142%
Subject 4	0.0481	0.0481	0.0481	0.0481	0.0481	0.0481	3.74e-5	0.078%
Subject 5	0.0436	0.0434	0.0440	0.0432	0.0440	0.0436	3.58e-4	0.820%
Subject 6	0.0416	0.0417	0.0417	0.0417	0.0417	0.0417	4.47e-5	0.107%
SAME-ECOS MWF MAEs								
Subject 1	0.0346	0.0345	0.0346	0.0346	0.0346	0.0346	4.47e-5	0.129%
Subject 2	0.0299	0.0300	0.0298	0.0300	0.0299	0.0299	8.36e-5	0.280%
Subject 3	0.0301	0.0301	0.0301	0.0301	0.0301	0.0301	1.73e-5	0.058%
Subject 4	0.0437	0.0435	0.0435	0.0436	0.0437	0.0436	1.00e-4	0.229%
Subject 5	0.0354	0.0353	0.0353	0.0350	0.0353	0.0353	1.52e-4	0.430%
Subject 6	0.0350	0.0351	0.0350	0.0352	0.0351	0.0351	8.37e-5	0.239%

Note: In each trial, simulated MR noise (SNR = 100) is added to the brain decay data of the six subjects, and the MAEs in MWF due to the added noise are calculated for NNLS and SAME-ECOS. The SD and coefficient of variance (CoV) of MAEs are also presented.

other quantities as an additional variable in the NNLS analysis. Another example would be how easily the influence of B1 inhomogeneity is handled. SAME-ECOS simply treats the refocusing FA as a variable when producing the simulated decay data to account for the FA variations caused by B1 inhomogeneity. However, the same problem was not solved using NNLS for many years until Prasloski et al¹⁷ proposed to integrate the stimulated echo correction into the original NNLS algorithm, which was a tremendous effort for such integration. Furthermore, if one wants to subject the data analysis to a simpler three-pool model for instance, SAME-ECOS can easily be simplified to a 3-pool solver by fixing the number of T_2 components to be $n = 3$ in the simulation, although it is not encouraged to do so for reasons discussed later. Conclusively, SAME-ECOS is extremely tunable to accommodate either a simpler or more complex model.

Another advantage of SAME-ECOS is to use the concept of resolution limit to prevent overfitting of noise. Due to unavoidable noise contamination, T_2 components can only be recovered to a certain extent, and any violation to the law of resolution limit should be theoretically prohibited. The essential part of SAME-ECOS training data production is to obey the resolution limit to prevent any solutions with adjacent T_2 components being too close together, a feature which is not guaranteed in the solutions of NNLS or any other methods to our knowledge. One reasonable concern that is raised here is that nature does not know about resolution limits, so what happens when adjacent T_2 components really are too close together? Unfortunately, the answer is that we may never resolve

these two peaks in a solution where they are bound to degenerate into one peak due to the limited SNR, as Istratov et al¹² pointed out about this particular problem: “Many physicists have discovered after much wasted effort that it is essential to understand the ill-conditioned nature of the problem before attempting to compute solutions”. On the other hand, the commonly used NNLS method has incorporated regularization techniques⁵⁰ to mitigate the noise overfitting, but the choice of regularization parameters is poorly justified in most of the MR literature. From a technical perspective, the strength of regularization should be selected based on the local SNR rather than being universally fixed. Unfortunately, this has never been practiced in the NNLS analysis due to technical challenges. In addition, the noise of a magnitude MR image originates from Gaussian distributed noise on both real and imaginary channels, which is not accounted for in the NNLS method⁵¹ but is correctly modelled in the SAME-ECOS approach. Particularly in the last few echoes, when there are no residuals of MR signal (only pure noise), the noise profile would follow a Rayleigh distribution on the magnitude image that always has positive values, such that NNLS could misinterpret the Rayleigh noise as a long T_2 component that has not been completely decayed yet. This phenomenon is observed in Figure 6, where NNLS mean spectra (red) for all ROIs always have a little rise on the far side of long T_2 s ($T_2 = 2000$ ms), but SAME-ECOS mean spectra (blue) is free of this problem.

SAME-ECOS also takes advantage of the strong predictive power of a fine-tuned neural network. For regression problems such as our T_2 -fitting task, modern ML methods

like neural networks usually outperform conventional statistical methods in terms of better prediction accuracy and faster data processing speed,²⁸ which is exactly what we have observed in our simulation tests. Out of various ML methods, the neural network approach is favored for use in the current SAME-ECOS workflow because it has been successfully implemented in similar tasks by different research groups.^{26,27,52} In theory, other ML methods may also achieve a similar predictive power. Comparisons between different ML methods are beyond the scope of this paper, but could be an area of future investigation.

Finally, it is worth highlighting that the simulated training dataset of SAME-ECOS is not informed by any prior knowledge (eg, a two-pool model is informed by the T_2 times of myelin water and IE water, or a typical range of the MWF values obtained from previous studies). Instead, SAME-ECOS is completely driven by a large number of random simulations from uniform distributions, which are only regulated by the experimental conditions such as SNR. This approach should be valued and favored because it is immune to (1) the potential errors of previous findings where the prior information is acquired from; and (2) the biases that prior information may introduce into the analysis results. Simply speaking, an informed model is more likely to perform the analysis with a bias naturally towards the prior information. Similar to NNLS, no prior information being needed is a desirable and a specially designed feature of SAME-ECOS.

5.4 | Disadvantages of SAME-ECOS

A major concern of SAME-ECOS is that its fitting results are not easily verified mathematically due to the use of a neural network. SAME-ECOS is capable of producing reliable results only on data that are similarly distributed as the training data. SAME-ECOS may become unpredictable and yield uninterpretable results when applied to unfamiliar data. Although we have generated a large training dataset by the simulations to account for many types of variations, there is no doubt that real experimental data have far more complexity in them. Factors such as artifacts are detrimental to any analysis methods, but SAME-ECOS is potentially less predictable. Unlike SAME-ECOS which empirically analyzes the data, conventional methods process the data either analytically or numerically, making them somewhat more mathematically explainable and predictable when encountering these problems.

Another limitation of SAME-ECOS is related to standardization. The variety of parameters such as the T_2 range, the noise profiles, the number of neural network hidden layers etc. can be manually selected at the user's will, which offers flexibility and customizability, but, at

this point, it is difficult to propose a standard SAME-ECOS model that works universally. The effects of changing these parameters warrants further investigation. Although SAME-ECOS demonstrated excellent performance in our simulation tests and in vivo example application, users should further validate SAME-ECOS on their own experimental data before replacing any conventional analysis methods.

5.5 | Other quantitative MRI techniques and beyond

Within MRI, the usage of SAME-ECOS is not just limited to multi-echo relaxation sequences. As long as the spins are trackable by simulations, then the SAME-ECOS methodology should also be applicable to other quantitative MRI techniques, such as multi-component driven equilibrium single pulse observation of T_1 and T_2 (mcDESPOT)⁵³ and neurite orientation dispersion and density imaging (NODDI),⁵⁴ by modifying the current workflow accordingly. Beyond MRI, it is also possible to apply SAME-ECOS to any quantitative sciences that involve multi-exponential decays. In general, we believe a data-driven solver like SAME-ECOS is an alternative way to produce at least comparable results to conventional methods. It may deliver better performance, especially when analytical and numerical solutions start to fail due to factors such as a limited amount of data points and low SNR. Nevertheless, the SAME-ECOS methodology seems to have the potential to be generalized for the analysis of decay data within and beyond the MRI field.

6 | CONCLUSIONS

We have introduced a novel method SAME-ECOS, which can decompose multi-exponential MR relaxation data into a T_2 spectrum. SAME-ECOS is highly tailorable to different experimental conditions and various analysis models. Compared with the commonly used method NNLS, our results have demonstrated that SAME-ECOS can yield much more reliable T_2 spectra and MWF values in a dramatically shorter processing time, by using the resolution limit constraint and neural network algorithm simultaneously.

ACKNOWLEDGMENTS

We thank the study participants and the excellent MRI technologists at UBC MRI Research Center. Funding support was provided by the Multiple Sclerosis Society of Canada, a Natural Sciences and Engineering Research Council of Canada Discovery Grant and an International

Collaboration on Repair Discoveries Seed Grant. HL was funded by an endMS Doctoral Studentship Award from the Multiple Sclerosis Society of Canada. TSJ was supported by a UBC MS Connect Summer Studentship Award funded by the Christopher Foundation.

DATA AVAILABILITY STATEMENT

The SAME-ECOS code is available at this URL: <https://github.com/hanwencat/SAME-ECOS>.

ORCID

Hanwen Liu  <https://orcid.org/0000-0002-5052-1424>

Piotr Kozłowski  <https://orcid.org/0000-0002-6462-2891>

Cornelia Laule  <https://orcid.org/0000-0002-9785-6918>

REFERENCES

- Whittall KP, MacKay AL, Li DK. Are mono-exponential fits to a few echoes sufficient to determine T2 relaxation for in vivo human brain? *Magn Reson Med*. 1999;41:1255-1257.
- MacKay A, Whittall K, Adler J, Li D, Paty D, Graeb D. In vivo visualization of myelin water in brain by magnetic resonance. *Magn Reson Med*. 1994;31:673-677.
- Laule C, Vavasour IM, Zhao Y, et al. Two-year study of cervical cord volume and myelin water in primary progressive multiple sclerosis. *Mult Scler*. 2010;16:670-677.
- Vargas WS, Monohan E, Pandya S, et al. Measuring longitudinal myelin water fraction in new multiple sclerosis lesions. *Neuroimage Clin*. 2015;9:369-375.
- Lang DJM, Yip E, MacKay AL, et al. 48 echo T(2) myelin imaging of white matter in first-episode schizophrenia: evidence for aberrant myelination. *Neuroimage Clin*. 2014;6:408-414.
- Baumeister TR, Kim JL, Zhu M, McKeown MJ. White matter myelin profiles linked to clinical subtypes of Parkinson's disease. *J Magn Reson Imaging*. 2019;50:164-174.
- Borich MR, Mackay AL, Vavasour IM, Rauscher A, Boyd LA. Evaluation of white matter myelin water fraction in chronic stroke. *Neuroimage Clin*. 2013;2:569-580.
- Wright AD, Jarrett M, Vavasour I, et al. Myelin water fraction is transiently reduced after a single mild traumatic brain injury—a prospective cohort study in collegiate hockey players. *PLoS One*. 2016;11:e0150215.
- Liu H, MacMillian EL, Jutzeler CR, et al. Assessing structure and function of myelin in cervical spondylotic myelopathy: evidence of demyelination. *Neurology*. 2017;89:602-610.
- Sharafi A, Chang G, Regatte RR. Biexponential T2 relaxation estimation of human knee cartilage in vivo at 3T. *J Magn Reson Imaging*. 2018;47:809-819.
- Sabouri S, Chang SD, Savdie R, et al. Luminal water imaging: a new MR imaging T2 mapping technique for prostate cancer diagnosis. *Radiology*. 2017;284:451-459.
- Istratov AA, Vyvenko OF. Exponential analysis in physical phenomena. *Rev Sci Instrum*. 1999;70:1233-1257.
- Whittall KP, MacKay AL. Quantitative interpretation of NMR relaxation data. *J Magn Reson*. 1989;84:134-152.
- Provencher SW. CONTIN: a general purpose constrained regularization program for inverting noisy linear algebraic and integral equations. *Comput Phys Commun*. 1982;27:229-242.
- Kroeker RM, Mark HR. Analysis of biological NMR relaxation data with continuous distributions of relaxation times. *J Magn Reson*. 1986;69:218-235.
- Lawson CL, Hanson RJ. *Solving Least Squares Problems*. Englewood Cliffs, New Jersey: SIAM. 1995.
- Prasloski T, Mädler B, Xiang Q, MacKay A, Jones C. Applications of stimulated echo correction to multicomponent T2 analysis. *Magn Reson Med*. 2012;67:1803-1814.
- Yoo Y, Prasloski T, Vavasour I, et al. Fast computation of myelin maps from MRI T(2) relaxation data using multicore CPU and graphics card parallelization. *J Magn Reson Imaging*. 2015;41:700-707.
- Doucette J, Kames C, Rauscher A. DECAES – DEcomposition and component analysis of exponential signals. *Zeitschrift für Medizinische Physik*. 2020;30:271-278.
- Du YP, Chu R, Hwang D, et al. Fast multislice mapping of the myelin water fraction using multicompartment analysis of T2* decay at 3T: a preliminary postmortem study. *Magn Reson Med*. 2007;58:865-870.
- Akhondi-Asl A, Afacan O, Mulkern RV, Warfield SK. T(2)-relaxometry for myelin water fraction extraction using wald distribution and extended phase graph. *Med Image Comput Assist Interv*. 2014;17:145-152.
- Bertero M, Boccacci P, Pike ER. On the recovery and resolution of exponential relaxation rates from experimental data: a singular-value analysis of the laplace transform inversion in the presence of noise. *Proc Math Phys Eng Sci*. 1982;383:15-29.
- McWhirter JG, Pike ER. On the numerical inversion of the Laplace transform and similar Fredholm integral equations of the first kind. *J Phys A: Math Gen*. 1978;11:1729-1745.
- Link N, Bauer S, Ploss B. Analysis of signals from superposed relaxation processes. *J Appl Phys*. 1991;69:2759-2767.
- Lundervold AS, Lundervold A. An overview of deep learning in medical imaging focusing on MRI. *Zeitschrift für Medizinische Physik*. 2019;29:102-127.
- Lee J, Lee D, Choi JY, Shin D, Shin H, Lee J. Artificial neural network for myelin water imaging. *Magn Reson Med*. 2020;83:1875-1883.
- Liu H, Xiang Q-S, Tam R, et al. Myelin water imaging data analysis in less than one minute. *Neuroimage*. 2020;210:116551.
- Schmidhuber J. Deep learning in neural networks: an overview. *Neural Networks*. 2015;61:85-117.
- Nagtegaal M, Koken P, Amthor T, et al. Myelin water imaging from multi-echo T2 MR relaxometry data using a joint sparsity constraint. *Neuroimage*. 2020;219:117014.
- Kumar D, Hariharan H, Faizy TD, et al. Using 3D spatial correlations to improve the noise robustness of multi component analysis of 3D multi echo quantitative T2 relaxometry data. *Neuroimage*. 2018;178:583-601.
- Does MD, Olesen JL, Harkins KD, et al. Evaluation of principal component analysis image denoising on multi-exponential MRI relaxometry. *Magn Reson Med*. 2019;81:3503-3514.
- Prasloski T, Rauscher A, MacKay AL, et al. Rapid whole cerebrum myelin water imaging using a 3D GRASE sequence. *Neuroimage*. 2012;63:533-539.
- Hennig J. Multiecho imaging sequences with low refocusing flip angles. *J Magn Reson*. 1988;78:397-407.
- Cárdenas-Blanco A, Tejos C, Irrarrazaval P, Cameron I. Noise in magnitude magnetic resonance images. *Concepts Magn Reson*. 2008;32A:409-416.

35. Klambauer G, Unterthiner T, Mayr A, Hochreiter S. Self-normalizing neural networks. *arXiv preprint arXiv:1706.02515*. 2017.
36. Bishop CM. *Pattern Recognition and Machine Learning*. New York: Springer; 2006.
37. Kingma DP, Ba J. Adam: a method for stochastic optimization. *arXiv*. 2014;1412.6980.
38. LeCun Y, Bengio Y, Hinton G. Deep learning. *Nature*. 2015;521:436-444.
39. Abadi M, Agarwal A, Barham P, et al. Tensorflow: large-scale machine learning on heterogeneous distributed systems. *arXiv preprint arXiv:1603.04467*. 2016.
40. Dvorak AV, Ljungberg E, Vavasour IM, et al. Rapid myelin water imaging for the assessment of cervical spinal cord myelin damage. *Neuroimage Clin*. 2019;23:101896.
41. Lee LE, Ljungberg E, Shin D, et al. Inter-vendor reproducibility of myelin water imaging using a 3D gradient and spin echo sequence. *Front Neurosci*. 2018;12:854.
42. Liu H, Rubino C, Dvorak AV, et al. Myelin water atlas: a template for myelin distribution in the brain. *J Neuroimaging*. 2019;29:699-706.
43. Liu H, Ljungberg E, Dvorak AV, et al. Myelin water fraction and intra/extracellular water geometric mean T2 normative atlases for the cervical spinal cord from 3T MRI. *J Neuroimaging*. 2020;30:50-57.
44. Ljungberg E, Vavasour I, Tam R, et al. Rapid myelin water imaging in human cervical spinal cord. *Magn Reson Med*. 2017;78:1482-1487.
45. Han J, Kamber M, Pei J. 2 - getting to know your data. *Data Mining (Third Edition)*. 2012;39-82.
46. Zhang Y, Brady M, Smith S. Segmentation of brain MR images through a hidden Markov random field model and the expectation-maximization algorithm. *IEEE Trans Med Imaging*. 2001;20:45-57.
47. Laule C, Kozlowski P, Leung E, Li DK, Mackay AL, Moore GR. Myelin water imaging of multiple sclerosis at 7 T: correlations with histopathology. *Neuroimage*. 2008;40:1575-1580.
48. Laule C, Leung E, Li DK, et al. Myelin water imaging in multiple sclerosis: quantitative correlations with histopathology. *Mult Scler*. 2006;12:747-753.
49. Celik H, Bouhrara M, Reiter DA, Fishbein KW, Spencer RG. Stabilization of the inverse Laplace transform of multiexponential decay through introduction of a second dimension. *J Magn Reson*. 2013;236:134-139.
50. Tikhonov AN, Goncharsky AV, Stepanov VV, Yagola AG. *Numerical Methods for the Solution of Ill-Posed Problems*. Vol. 328. Dordrecht: Springer Science & Business Media; 2013.
51. Bjarnason TA, McCreary CR, Dunn JF, Mitchell JR. Quantitative T2 analysis: the effects of noise, regularization, and multivoxel approaches. *Magn Reson Med*. 2010;63:212-217.
52. Cohen O, Zhu B, Rosen MS. MR fingerprinting deep Reconstruction Network (DRONE). *Magn Reson Med*. 2018;80:885-894.
53. Deoni SC, Rutt BK, Arun T, Pierpaoli C, Jones DK. Gleaning multicomponent T1 and T2 information from steady-state imaging data. *Magn Reson Med*. 2008;60:1372-1387.
54. Zhang H, Schneider T, Wheeler-Kingshott CA, Alexander DC. NODDI: practical in vivo neurite orientation dispersion and density imaging of the human brain. *Neuroimage*. 2012;61:1000-1016.

SUPPORTING INFORMATION

Additional Supporting Information may be found online in the Supporting Information section.

FIGURE S1 Comparison of the cosine similarity scores of SAME-ECOS with and without resolution limit (RL) using a simulation phantom. SAME-ECOS with RL has higher similarity than without RL in all scenarios

FIGURE S2 Comparison of the mean absolute error (MAE) in myelin water fraction (MWF) estimation of SAME-ECOS with and without resolution limit (RL) using a simulation phantom. SAME-ECOS with RL has lower MAEs than without RL in all scenarios

FIGURE S3 Comparison of the MWF maps of SAME-ECOS with and without resolution limit (RL) using in-vivo data. SAME-ECOS without RL produces noisier MWF maps compared with SAME-ECOS with RL

FIGURE S4 Comparison of the T2 spectra of SAME-ECOS with and without resolution limit (RL) using in-vivo data. For all ROIs, the mean spectra of SAME-ECOS with RL are able to distinctively resolve a short and small T2 peak (attributed to myelin water) in addition to a more dominant middle peak (attributed to IE water), whereas the mean spectra of SAME-ECOS without RL can only resolve a single broad middle peak that intrudes into the myelin water region

How to cite this article: Liu H, Joseph TS, Xiang Q-S, et al. A data-driven T₂ relaxation analysis approach for myelin water imaging: Spectrum analysis for multiple exponentials via experimental condition oriented simulation (SAME-ECOS). *Magn Reson Med*. 2022;87:915-931. <https://doi.org/10.1002/mrm.29000>